Leveraging probabilistic forecasts for dengue preparation and control in Brazil

Luiz Max Carvalho*

* with the Infodengue & Mosqlimate Research Teams

October 15, 2025

Building ensembles of predictive results for dengue in Brazil

Goals:

- The Infodengue project works closely with the Brazilian MOH to help manage arbovirus diseases, through its early warning system.
- The Mosqlimate project's mission: Facilitate modeling the impact of climate on the dynamics of Arboviruses
- The invited teams are collaborators of the Infodengue team with experience with dengue modelling.

Organized by: `mosqlimate.org` and `info.dengue.mat.br`

Participants

15 teams submitted 19 predictive models for dengue in Brazil.

D-fense: UERJ/LNCC (Brazil)

Dengue oracle: FGV-EMAp (Brazil)

LaCiD/UFRN : LaCiD-UFRN (Brazil)

DS_OKSTATE : OSU (USA)

Beat it : Fiocruz (Brazil)

Strange Attractors : FGV-EMAp (Brazil)

DengueSprint : Cornell University (USA)

TSMixer ZKI-PH4 : Robert Koch Institute (Germany)

CERI Forecasting Club : CERI Stellenbosch University (South Africa)

Imperial College London : ICL (UK)

GeoHealth: KAUST (Saudi Arabia)

Global Health Resilience: BSC (Spain)

ISI Foundation: ISI (Italy)

JBD-Mosqlimate: FGV EMAp (Brazil)

Preditores da Picada: IMPA-Tech (Brazil)

Models

Model ID	Model	Approach
108	Preditores da picada	Autoregressive
133	Chronos-Bolt	Machine learning / Neural Network
134	ISI_Dengue_Model	Mechanistic
135	GHR Model	Bayesian
136	Imperial-TFT Model	Machine learning / Neural Network
143	Strange Attractors	Bayesian
144	Beat it	Bayesian
145	CNNLSTM	Machine learning / Neural Network
150	LNCC-AR_p-1	Autoregressive
152	LNCC-CLIDENGO-1	Mechanistic
154	LNCC-SURGE-1	Mechanistic
155	Dengue Oracle M1	Machine learning / Neural Network
156	Dengue Oracle M2	Machine learning / Neural Network
157	UERJ-SARIMAX-2	Autoregressive
158	DengueSprint_Cornell-PEH_2	Bayesian

Scoring

The **interval score** for a probability level $\alpha \in (0, 1)$ is

$$\begin{aligned} \mathrm{IS}_\alpha(l, u; y) = u - l &+ \frac{2}{\alpha}(l - y)I\{y < l\} \\ &+ \frac{2}{\alpha}(y - u)I\{y > u\} \end{aligned} \tag{1}$$

where I is the indicator function, y is the observed value of an r.v. Y and (l, u) is an $\alpha \times 100\%$ prediction interval for Y. We score models with the **weighted interval score (WIS)**

$$\mathrm{WIS}_{\mathcal{A}}\left([l_a, u_a]_{a \in \mathcal{A}}; y\right) = \sum_{a \in \mathcal{A}} a \cdot \mathrm{IS}_a\left(l_a, u_a; y\right) \tag{2}$$

Forecast puzzle

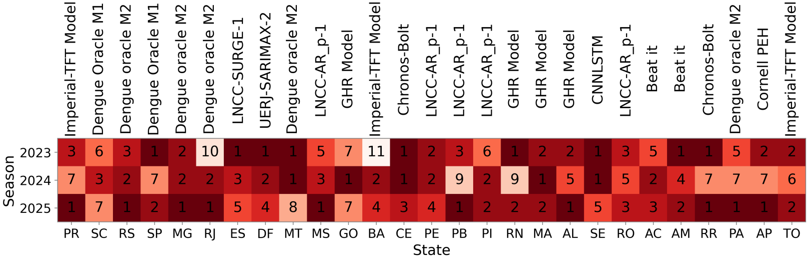


Figure 1: Ranking position of the best model throughout the entire period in each validation season. The ranking was calculated based on the WIS in the season.

Ensemble methods

All models submitted predictions consisting of the following percentiles $(p_{2.5}, p_5, p_{10}, p_{25}, p_{50}, p_{75}, p_{90}, p_{95}, p_{97.5})$.

Ensemble models

The ensemble models are built as the median of the model's predictive intervals. This approach was originally proposed by the COVID Hub [a].

[a] Ray et al. (2023) *International Journal of Forecasting*

The resulting models were similar to those obtained by logarithmic pooling with equal weights, used in the previous sprint, but this methodology does not require a parametric approximation.

Score Normalization

- For each validation test, a normalized WIS score was computed by aggregating across all weeks as defined below:

$$\text{WIS}^{\text{norm}} = \frac{1}{Y_{\text{total}}} \sum_{t=1}^{T} \text{WIS}_t, \tag{3}$$

where:

- WIS_t is the Weighted Interval Score for week t,
- Y_{total} is the total number of cases in the validation period, and
- T is the total number of weeks in the validation period.

An ensemble built from the **top 5** models from the first two validation sets, was compared to individual models, in 2025, in terms of relative Skill scores.

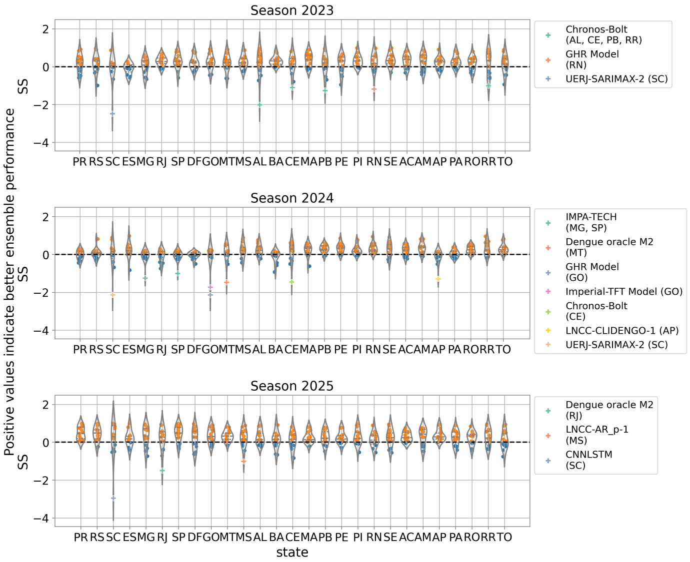
Ensemble (all models) vs. Individual Models

To assess the performance of the ensemble relative to the individual models, we computed the Skill Score (SS) of the Weighted Interval Score (WIS), defined as follows:

$$SS_{m,v} = 1 - \frac{WIS^{\text{norm}}_{\text{Ensemble},v}}{WIS^{\text{norm}}_{m,v}}, \tag{4}$$

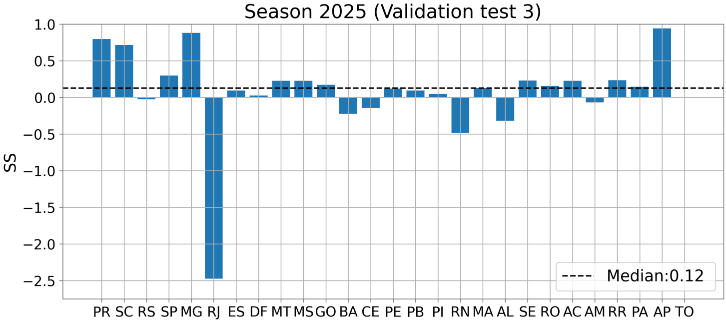
where m denotes an individual model and v corresponds to a specific validation test. **Positive values indicate that the ensemble model outperforms the individual model in terms of WIS.**

Ensemble (all models) vs. Individual models



Ensemble (top 5) vs. best model.

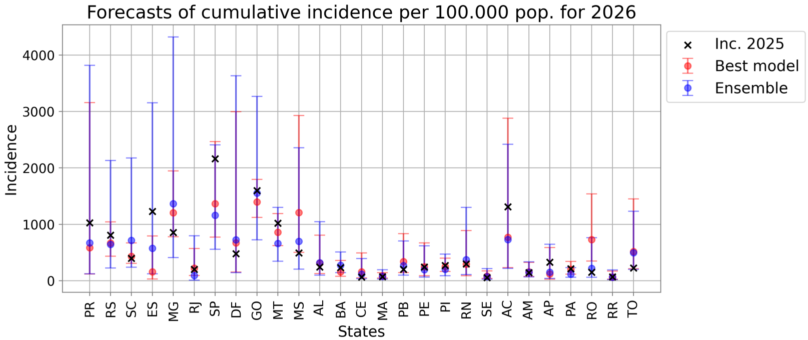
Ranked according to the mean WIS^{norm} at validations tests 1 and 2



This ensemble is more consistent across states and, on average, **12 % more accurate** than the best model.

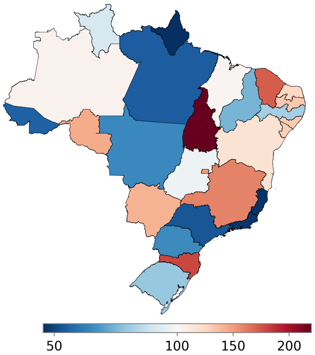
Brazil

Cumulative incidence:



Brazil (incidence map)

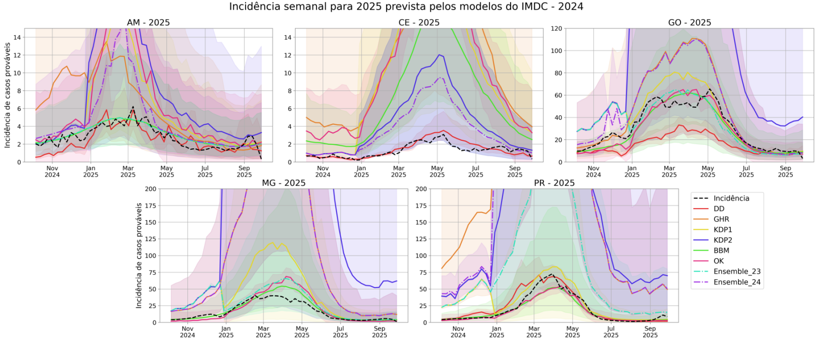
Inc. pred 2026/Inc. 2025 (%)



State	Inc. pred 2026	Inc. 2025	Ratio 2026/2025 (%)
PR	671.35	1025.61	65.46
RS	641.57	807.54	79.45
SC	716.06	398.09	179.87
ES	573.74	1228.34	46.71
MG	1364.69	856.06	159.42
RJ	93.80	200.45	46.80
SP	1158.15	2162.84	53.55
DF	726.88	477.70	152.16
GO	1553.60	1599.21	97.15
MT	661.82	1019.34	64.93
MS	696.60	492.99	141.30
AL	318.64	242.97	131.14
BA	273.53	233.19	117.30
CE	115.37	66.74	172.85
MA	75.33	73.29	102.79
PB	266.78	202.19	131.94
PE	196.34	241.37	81.34
PI	202.54	270.61	74.85
RN	373.56	298.75	125.04
SE	73.14	56.39	129.71
AC	728.84	1309.63	55.65
AM	150.99	143.65	105.11
AP	149.75	329.08	45.51
PA	116.80	212.92	54.86
RO	218.13	151.13	144.34
RR	60.95	67.10	90.83
TO	496.32	225.82	219.78

Overall model+ensemble ranking and predictions (2024 to 2025)

RANK	AM	CE	GO	MG	PR
1	BBM	DD	OK	BBM	OK
2	Ensemble_23	Ensemble_23	BBM	OK	BBM
3	DD	Ensemble_24	Ensemble_23	Ensemble_23	DD
4	Ensemble_24	KDP2	KDP1	KDP1	KDP1
5	KDP1	BBM	DD	Ensemble_24	Ensemble_23



Mosqlimate Team

Thanks! And see you again soon...

Checkout our report!